

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Concept Mapping

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
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Section / Batch:	B.Tech IT	Date:	10/07/2026

Code of Conduct

A. Gopika/192521343

I _____ (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 Concept mapping questions. Total: 100 Marks.
- When a student answer using a concept map, in evaluation the answer should be with following four requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Linked data structure	Basic data structure	1	20
Primitive and Non-Primitive Data Structure - Linear and Non-Linear Data Structure	Classifications of Linear data structure	2	20
Search Data Structure	Linear Search and Binary Search	3	20
Recursion, Performance analysis	Recursion	4	20
Tree Data Structures	Classifications of Tree data structure	5	20
GRAND TOTAL:		100 Marks	

Annexure A – Concept Mapping

1. A student attendance system records daily presence data. Construct a concept map linking Array → Fixed Size → Sequential Access → Linked List → Dynamic Allocation → Flexibility. Explain how each structure works and justify which is more suitable for dynamic data updates. (20 Marks)
2. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing. (20 Marks)
3. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique. (20 Marks)
4. A program processes nested expressions using recursion and may encounter deep nesting. Construct a concept map linking Recursion → Call Stack → Stack Overflow → Error Handling → Efficiency. Analyze potential issues and justify strategies to handle them. (20 Marks)
5. A company maintains hierarchical employee records with multiple levels of management. Construct a concept map connecting Tree → Nodes → Levels → Traversal (DFS/BFS) → Searching → Efficiency. Analyze how traversal techniques affect performance and justify the most suitable approach for retrieving employee data. (20 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____



STUDENT ATTENDANCE SYSTEM

ARRAY

FIXED
SIZE

SEQUENTIAL
ACCESS

LINKED
LIST



Stores data in
contiguous
memory

Size cannot
change after
creation

Elements are
accessed in
order

Nodes
Connected using
pointers

DYNAMIC ALLOCATION

⇒ Memory allocated when
needed

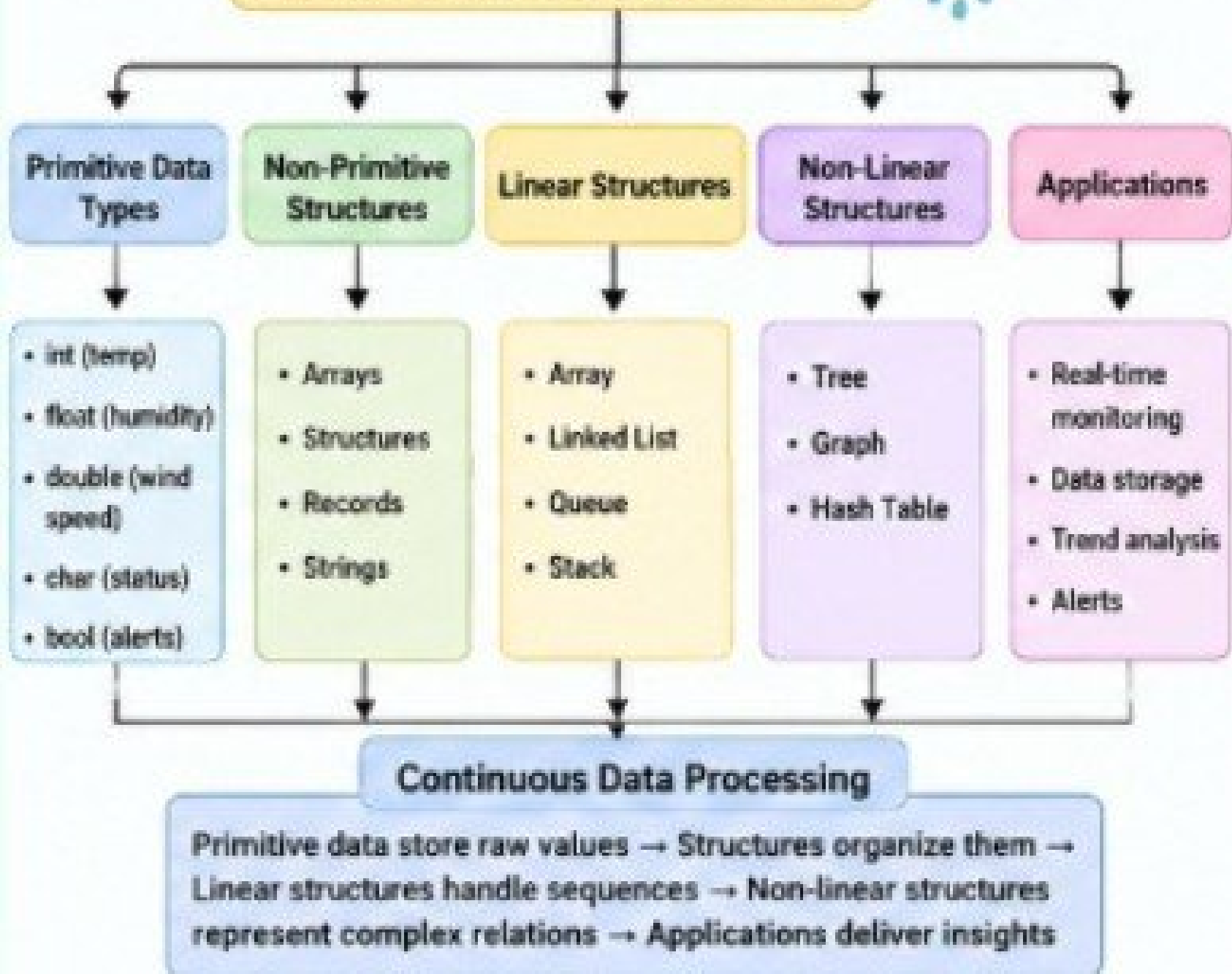
⇒ Nodes created
dynamically

Easy
insertion

Easy
Deletion

Efficient
Updates

Weather Monitoring System



Digital Archive (Searching)



Searching

Process of
finding a
particular
document

Linear Search

- Check one by one
- Works on any data
- Time Complexity $O(n)$

Binary Search

- Divide & conquer
- Works only on sorted data
- Time Complexity $O(\log n)$

Sorted Data

- Data arranged in order
- Essential for binary search

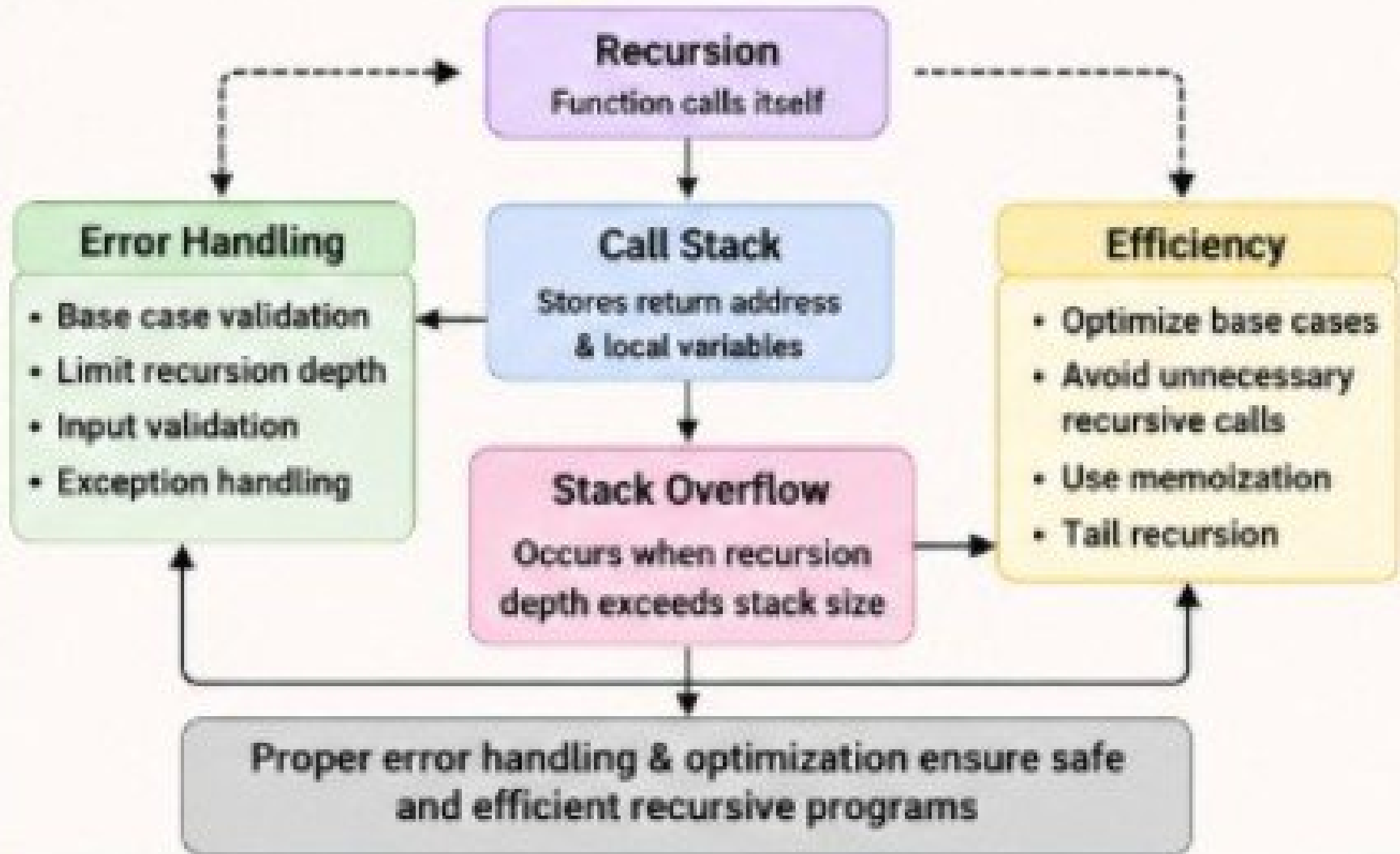
Time Complexity

Measures time taken
to search

Performance

Better performance
with sorted data
using Binary Search

Question 4



Question 5

Tree (Employee Hierarchy)



Nodes

- Represent employees
- Each node contains data

Levels

- Show hierarchy
- Root = CEO
- Children = Managers
- Leaf = Staff

Traversal (DFS / BFS)

- DFS (Depth First)
→ Explore deep first
- BFS (Breadth First)
→ Level by level

Searching

- Find employee quickly
- Depends on traversal & structure

Efficiency

- DFS → Less memory (stack)
- BFS → Better for level-wise data

BFS is more suitable for hierarchical employee retrieval (level-wise access), while DFS is useful for deep hierarchy exploration.